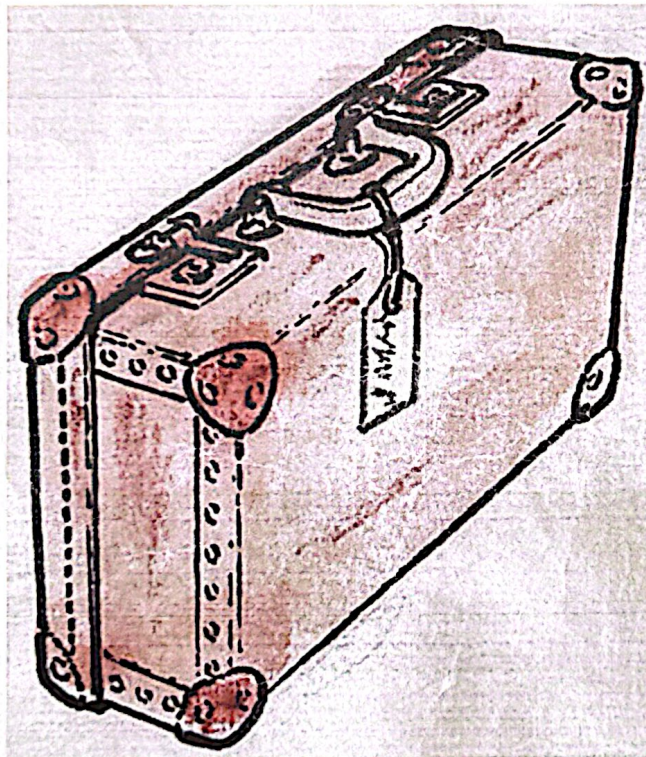


2025 Practical exam case studies

Clinical Reasoning in Musculoskeletal Rehabilitation

UZYKDK-15-2

These are the 4 seen case studies for your exam - We will work through these over the next few days - Please remember to bring to every session!



Tom

C₁-C₂. C₃. C₄. C₅. C₆,
C₇

8: C₂ 7/12

L₁. L₂. L₃. L₄. L₅

T₁ → T₁₂

Case Study 1

Thoughts?

Patient

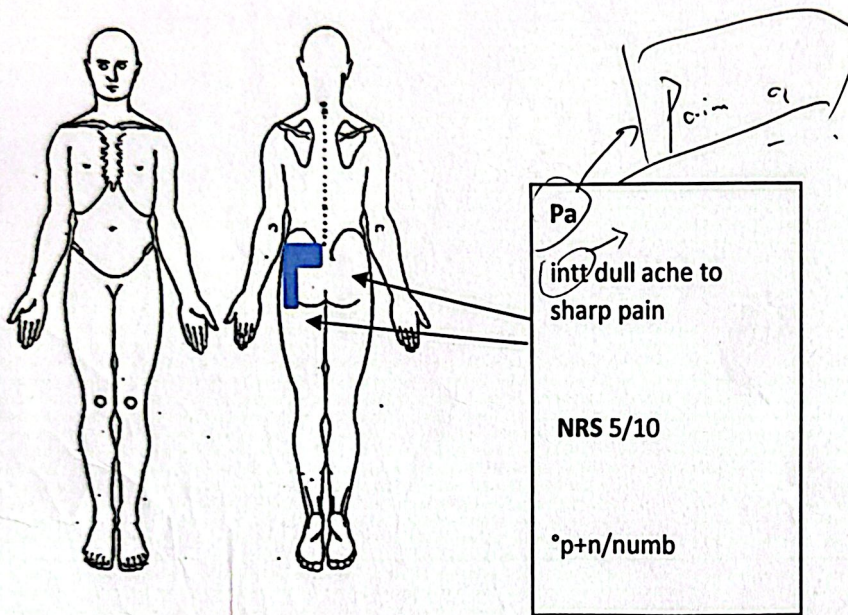
Female, aged 37

SUBJECTIVE

Main problem: Pain over the left buttock

Affecting childcare & mobility

Body chart



Aggravating factors

Lifting /carrying baby – immediate ↑ Pa

Getting dressed – reaching to put on socks – immediate Pa

Walking – esp downhill > 10mins Pa

Standing – on Left LL

Sitting in midline for > 30 minutes – Pa – worse if leaning to left

Easing factors

Lying on left side with hip and knee flexed to 90° ~ 30 mins

Supine lying with hips and knee flexed in crook lying supported.

Night time

Can wake up if lying on right side. Forced to sleep on left side only (not normal for her).

A.M

Generally fine

24 hour pattern

Activity dependent. No morning joint stiffness

Special questions

Bladder – mild urge incontinence since birth of first child, non-progressive. No change since her MRI.

Bowel control/saddle sensation/sexual function: NAD

*p+n/*numb

Gait – some discomfort when WB as per Aggs *trips/*LL weakness

*fever/*fatigue/*wtloss/*night sweats

Thoughts on special questions/ red flags?

History of present condition

Gradual onset of pelvic girdle pain during pregnancy. Pain continued after birth of 2nd child (2 years ago). Has not seen any professional since midwife, just got on with things. Increased concern over persistent nature of symptoms. GP has recently referred for investigations – all normal. Her condition is not worsening.

Previous treatments include a belt supplied by midwife and given pelvic floor exercises to do.

Occasional use of pain relief.

Past medical history

Fit and well

*Thyroid/*Heart/*BP/*Inflammatory Arthritis/*Asthma

*Epilepsy/*Diabetes/*Surgery/*Cancer

Thoughts on HPC/PMH?

Drug history

Contraceptive pill

*Long term steroids, *Anticoagulants

Investigations

Lumbar/pelvis x-ray – NAD

Lumbar MRI – NAD

Blood tests – NAD

Social history

Lives with wife and 2 children (2,4)

This is affecting her childcare & mobility

Occupation

Nurse part time. Works on a trauma and orthopaedic ward.

Hobbies

Fitness classes - Pilates and Yoga – stopped since children. No time/unsure whether to go as worried could make her condition worse. Keen to get fit again

General mood

Gets down because of her pain disorder but doesn't feel she is depressed. Has never been depressed.

Notices the pain increases when she feels stressed looking after child / not sleeping

Beliefs/concerns

Unsure of whether to exercise. Doesn't know what will help

Has 'no time'

Has been told it's her SIJ that's "out" but equally confused as to why she has pain and has a 'normal' MRI. Worried the pain could be all in her head now.

Expectations

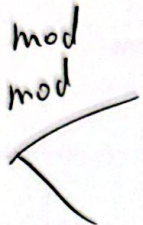
Put her SIJ back in place?

Help to restart her exercise classes

Thoughts on any yellow flags? →

ANALYSIS

Initial Hypothesis/ Clinical diagnosis:

S mod
I mod
N  myogenic: glut mgn // hip flex
 transverse
 Aerogenic: → hip joint
 psycological: Wound about SIJ back in place // rest exercise classes

been depressed

Any precautions/contraindications for Ax?

Differential diagnosis / alternative hypotheses?

OBJECTIVE

Observation

Slight increase in lumbar lordosis.

Stands with ↓ WB on left, with Left hip in slight flexion.

Lsp ROM

	AROM	Active over pressure
Flex*	Fingertips to ankle jt – No pain on flex, but pain on returning to neutral	√
Ext	Finger tips to just below buttock crease Pa	N/a
L SF	Finger tips to mid thigh Pa	N/a
R SF	√	Slight discomfort EOR

Special tests

Hip joint quadrant: NAD

SIJ tests:

- 
- Ilium compression - 'relief'
 - *Thigh thrust - +ve on left Pa
 - SI joint 'gapping' - +ve
 - Prone sacral thrust - +ve Pa
 - *Gaenslen's test - +ve on left Pa

Leg length – 92cm ASIS to lat malleolus R=L

Active SLR – marked heaviness elevating Left leg and breath holding

Thoughts on special tests? →

Muscle / movement control

Pelvic floor contraction associated with bracing of abs and breath holding

TrA bracing/ breath holding – unable to sustain > 5 secs

↓ isometric glut med contraction on left 4/5 and fatigues quickly on repeated clam exercise 7 reps (Right ~18reps)

PAMs / palpation

PPIVMs – NAD

PAIVMS – PSIS discomfort or maybe soreness on palpation L SI jt???

tender trigger point piriformis / glutes

Thoughts on movement control and PAMS →

Functional tests

Sitting: slumped sitting weight shifted to right for comfort.

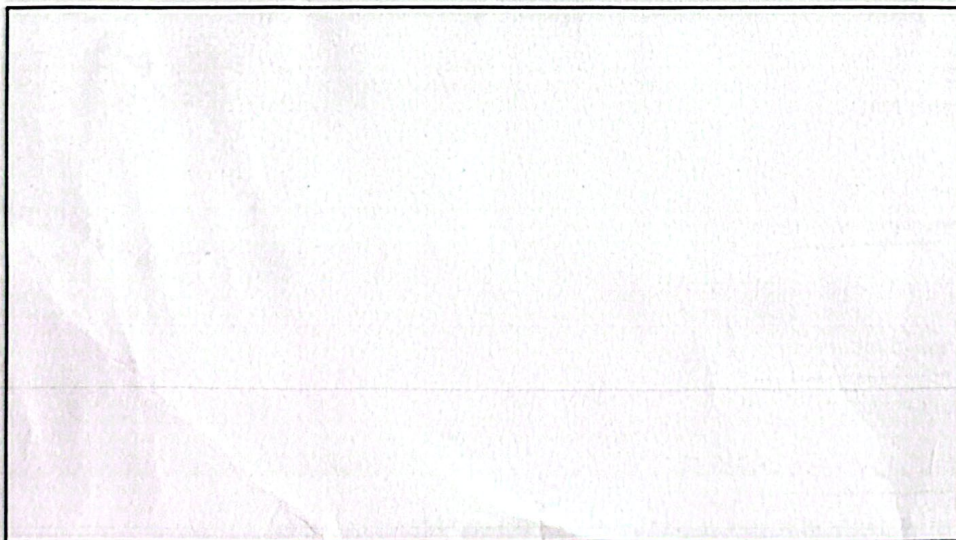
Sit to stand – tendency to shift to right and load on right leg

Single leg stand - ↑sway to left and Trendelenberg sign/pattern

Single leg sit to stand – unable left leg - Pa

Thoughts on function →

ANALYSIS



Hypothesis/ Clinical impression:

PCP
~~pelvic~~ girdle pain (骨盆疼痛)
骨盆周围韧带和关节的疾病

Clinical Reasoning (Why)?

Problem list / Contributing Factors.

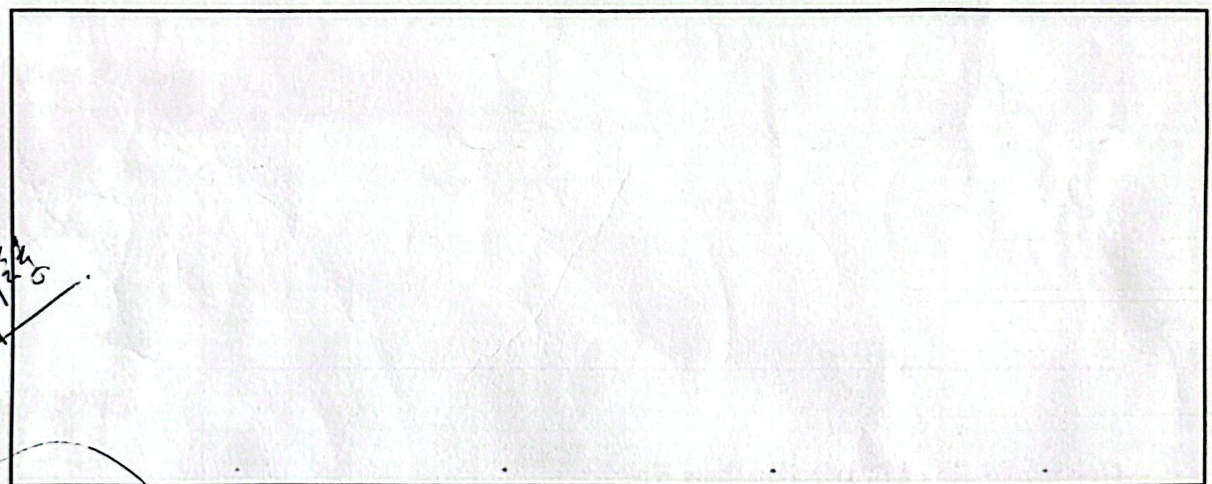
pelvic girdle pain
pelvis discomfort.

- ① weak gluteus medius
- ② Pain
- ③ WT Bending
- ④ Worried to exercise
- ⑤ weak core.

Treatment plan

- ① Heat therapy (increase metabolism) ~~not recommended~~ } reduce pain
- ② Trigger point: ^{piriformis} _{iliopsoas} the muscle relax,
- ③ ~~Dead bug~~ Strengthen Glut medius $\Rightarrow$ side abduction with band
- ④ Stretch Glut medius $\Rightarrow$ lunge steps
- ⑤ Deadbug with Ball to hold (easier if it is too hard) $\Rightarrow$ core
- ⑥ posture education & encouragement.

Clinical Reasoning – Why??



Goals

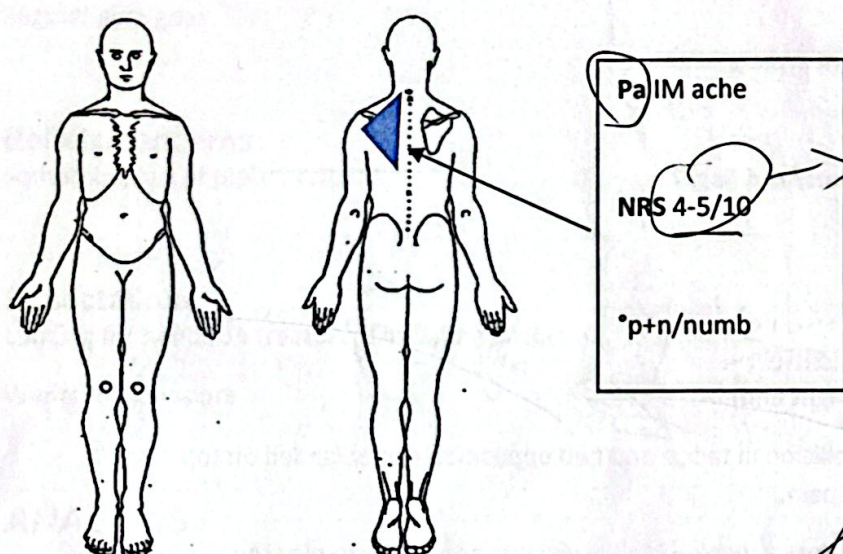
- S : restful exercise class
- MA : reduce the pain 5/10 - 0/10 1 month.
- A : progress or add up more exercise
- I :
- T :

Thoughts?

SUBJECTIVE

Localised thoracic pain around medial border left scapula

Body chart



Twisting movements – occasional Pa

Sitting on sofa – 30mins Pa

Sitting at desk at work - 60 mins Pa

Rotating spine

Pain on deep breath or cough - eases one minute

Hot pack

Changing position, getting up from sitting – 10-15 minutes

$$S: \text{mod}$$

N: muscl. i
up fr.
 Nervus $\rightarrow$ do slumb
to claws
lack of rage
 Asthroya $\rightarrow$ Exempt
pro. symmetrical
Shrubs
vertical
fr. vol.

Can't play game
weekend

24 hour pattern

Night time

Occasionally wakes the patient 1-2x /week. Can get back to sleep

A.M.

Pa increases throughout the day, depends on how much he is sitting, and whether he is working from home or at work.

Special questions

Occasional night pain – wakes but can get back to sleep

*p&n/*numb/*gait/*balance/*trips/*LL weakness/*clumsiness

*D's/ N's

*fever/*fatigue/*wtloss/*night sweats

Cough/sneeze: NAD

Thoughts on special questions/ red flags? →

History of present condition

Plays American Football – (Tight End)

Game 2 months ago – Big collision in tackle and two opposition player landed on top of him on floor. Immediate pain.

He normally goes to gym 3 times a week – Mainly weights and wants to play this weekend but unsure if he should.

Past medical history

Previous appendectomy

*Thyroid/*Heart/*BP/*Inflammatory Arthritis/*Asthma *Epilepsy/*Diabetes/*Cancer

Thoughts on HPC/PMH? →

Drug history

Takes creatine supplement

*Long term steroids, *Anticoagulants

any investigation?

truma
(maybe)

Investigations

Nil

Social history

Single

Occupation

Work in finances – office based/work from home. Full time. Still works from home 2-3x a week.

Hobbies

American Football - Club level

Regular gym goer

Beliefs/concerns

Something out of place in ribs

Expectations

Looking for hands on treatment to fix the problem.

Wants to play sport

Yellow flags

ANALYSIS

Hypothesis/ Clinical diagnosis:

S

I

N

Any precautions/contraindications for Ax?

Differential diagnosis / alternative hypotheses

OBJECTIVE

Observation

Kyphotic upper thoracic posture. Increased lordosis mid Csp with forward head posture and protracted shoulders. Muscular build.

Slight increased prominence of the inferior angle of L scapula

Lsp ROM
Full AROM, pain

Thru c1 & spine

Tsp ROM

	AROM	Overpressure
Flex	3/4- full range, Pa	NT
Ext	1/2 range stiff	stiff
L Rotn	3/4 range, stiff	stiff
R Rotn	1/2 range, Pa	√√
L SF	3/4 range stiff	stiff
R SF	√√	Mild Pa

*Q, J, J Thru c1
↓
cervical
&
functional*

Csp ROM

→ Cervical spine

	AROM	Overpressure
Flex	√√	Mild Pa
Ext	√√	√√
L Rotn	√√	√√
R Rotn	√√	Mild Pa
L SF	√√	√√
R SF	√√	Mild Pa

Thoughts on movement tests? →

Shoulder

Full AROM, pain

Muscle tests

Upper traps short and 'tight' on L

Levator scapula short and tight on the L – tight Pa

Scapular setting – poor awareness, unable to hold 10 neutral posture.

Reduced proprioception / awareness of neutral spinal posture in sitting

Palpation /PAIVMs

Palpation – tender left paraspinal medial border scapular and upper traps Trp

PAIVMs - PA and unilateral T2/3 Pa, cervical spine NAD

Thoughts on muscle tests and palpation? →

Neuro

Dermatomes: NAD to light touch upper and lower limbs

Myotomes :L=R 5/5 C1-T1, L2-S1

Reflexes: Biceps, Triceps, Knee and Ankle jerk all equal and present, normal Babinski and Clonus

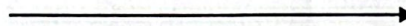
Thoughts on neuro tests



Functional tests

Sits with thoracic spine kyphosed++ and poking chin with protracted shoulder girdle.

Thoughts on functional tests



ANALYSIS

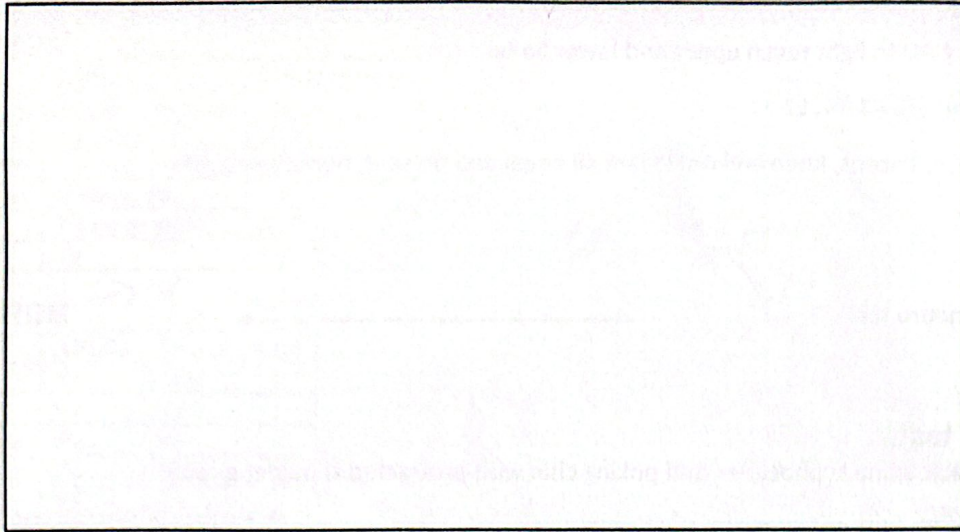
Other
back

slump

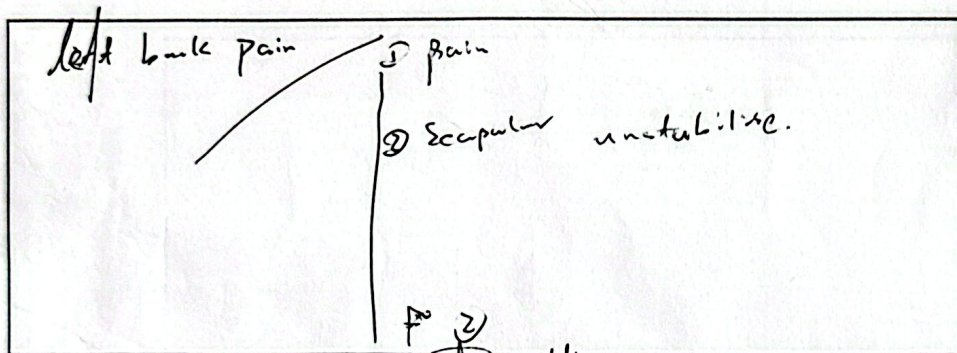
pee length

Hypothesis/ Clinical impression:

Clinical Reasoning (Why)?



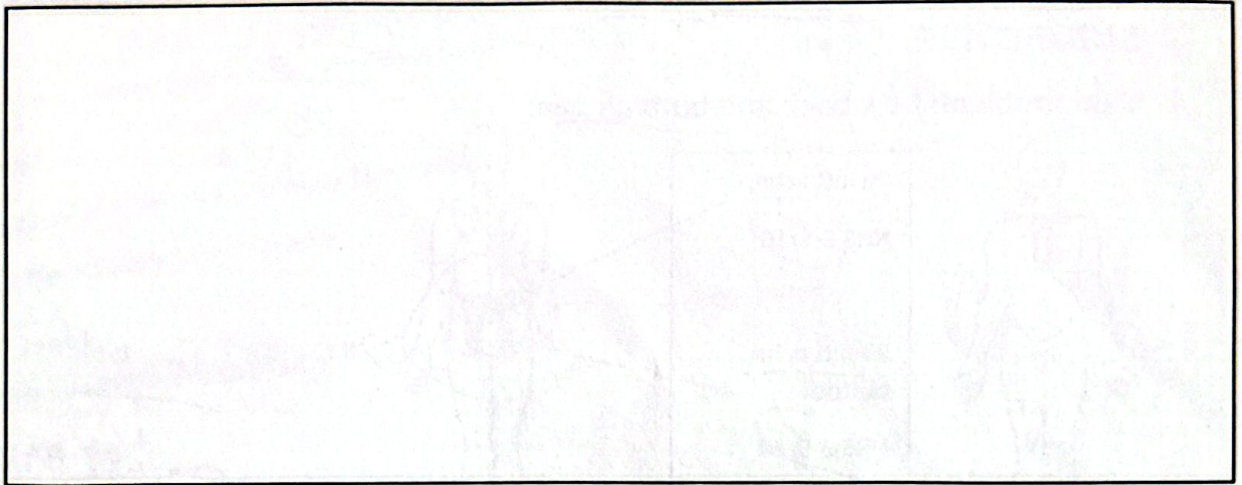
Problem list / Contributing Factors.



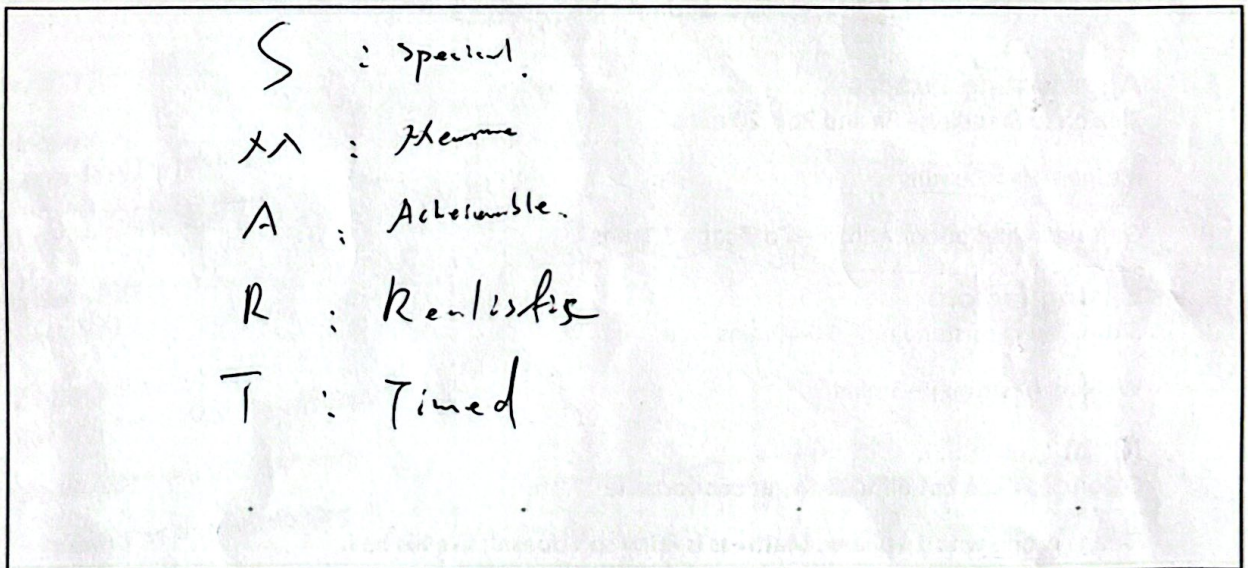
Treatment plan

- ① Scapular head therapy (mobilise) rotator tissue.
- ② PAZVM: Thoracic spine & facet joint (relief joint?)
- ③ trigger point: left ~~right~~ muscle. (release sore point)
=> (pec, minor, minor, & rhombus?)
- ④ Threadle exercise: improve ROM of thoracic spine => Side bell game. (左右接球)
- ⑤ YTW & ~~side weight~~ "fly" exercise (light? straight?) => strength & proprio.
- ⑥ Wall strength => pectoralis minor.
- ⑦ Education the posture. < ⑧ brush nerve.

Clinical Reasoning – Why??



Goals

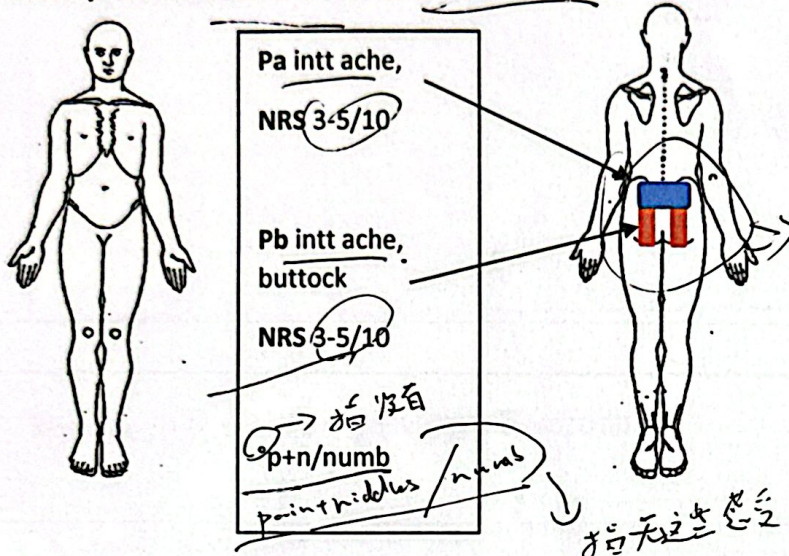


Case study 3 Patient: Male, aged 23

Thoughts?

SUBJECTIVE

Main problem: Low back and buttock pain



Aggravating factors

Flexion in Standing - Pa and Pb > 20 mins

Sitting - Pa > 30mins

Getting in and out of a chair - Pa if sat > 10mins

Easing factors

Sitting/lying (if standing) - 30-40mins

Walking (if sitting) - 5mins

Night

OK once asleep but difficult to get comfortable

Sleeps prone with 1 +pillow. Mattress is fairly soft doesn't like his bed.

Am

Early morning stiffness 10mins. Ok once had a cup of tea and down stairs.

History of present condition

Weightlifter - 3 months ago Deadlifting 200kg and felt something 'go' in lower back.

Immediate pain - Intermittent and variable since

Past medical history

Mild asthmatic

*Thyroid/*Heart/*BP/*Inflammatory Arthritis/*Epilepsy/*Diabetes/*Surgery/*Cancer

Drug history

Co-codamol – uses as and when

复方镇痛药

Paracetamol
codeine

Inhalers – Blue and brown

吸入器

*Long term steroids, *Anticoagulants

抗凝药

Special questions

Bladder/bowel control/saddle sensation/sexual function: NAD

长期服用皮质激素

*p+n/*numb

No Ab normal Detected.

Gait – discomfort when WB but *trips/*LL weakness

*fever/*fatigue/*wtloss/*night sweats

Thoughts on special questions/ red flags?

Investigations

None – Wants MRI scan

Social history

Secretary

Lives with parents

Hobbies

Trains 5 days / week in gym

Mainly weight lifting

Beliefs/concerns

Coach said it could be a disc problem

Expectations

Advice / help. Thinks needs scan

Upset that can't train properly

Thoughts on any yellow flags?

ANALYSIS

Hypothesis/ Clinical diagnosis:

S

I

N

Any precautions/contraindications for Ax?

mod

mechanical

— poss nerve

— poss LV disc

has 3-5 probs => mild-mod.

OBJECTIVE

Observation

Muscular build. Is in spinal balance with lumbar and thoracic curves within normal limits

Thoughts on postural findings? →

Lsp ROM

	AROM	Overpressure
Flex	Finger tips to mid shin, Pa almost immediate	N/A
Ext*	Upper thigh Pa	Stiff
L SF	Mid shin ✓	✓
R SF	Mid shin ✓	✓

Special tests

Hip Quadrant: VV

SIJ: VV

Sacro iliac joint

Thoughts on movement tests? →

Muscle tests

Bridge – weak – unable to hold > 5 secs →

Neuro

Dermatomes: L=R, light touch

Myotomes: L2-S1, L=R, 5/5

Reflexes: Knee and ankle jerk, equal and present

Babinski and clonus: -ve

Neuro dynamic tests

SLR: 70 bilat – Buttock pain with +ve sensitisation bilaterally

Thoughts on neuro findings? →

Palpation & PAIVMs

↓ T10-L5 - tenderness over L4/5 Pa

Thoughts on palpation findings? →

*Neuro - Neurogenic
muscle - myogenic
joint - arthropathic.*

No red flags

straight leg raise

Function

Sit to stand – Normal

Step up and down – 'Trendelenburg' sign

Thoughts on function findings? →

坐立时 → 臀肌无力 → 臀肌无力

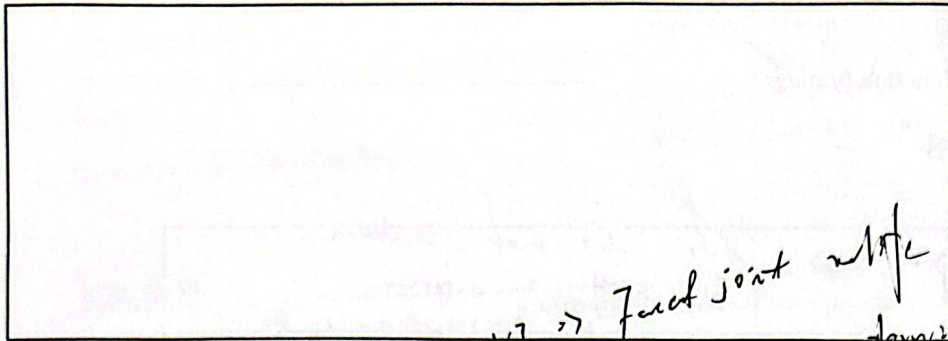
ANALYSIS

- problem =>
- (1) glute med weakness
 - (2) stiff into extension
 - (3) Neural interference - piriforms.
 - (4) Flexion control.
 - (5) unable to train
 - (6) nerves / various about "disc"

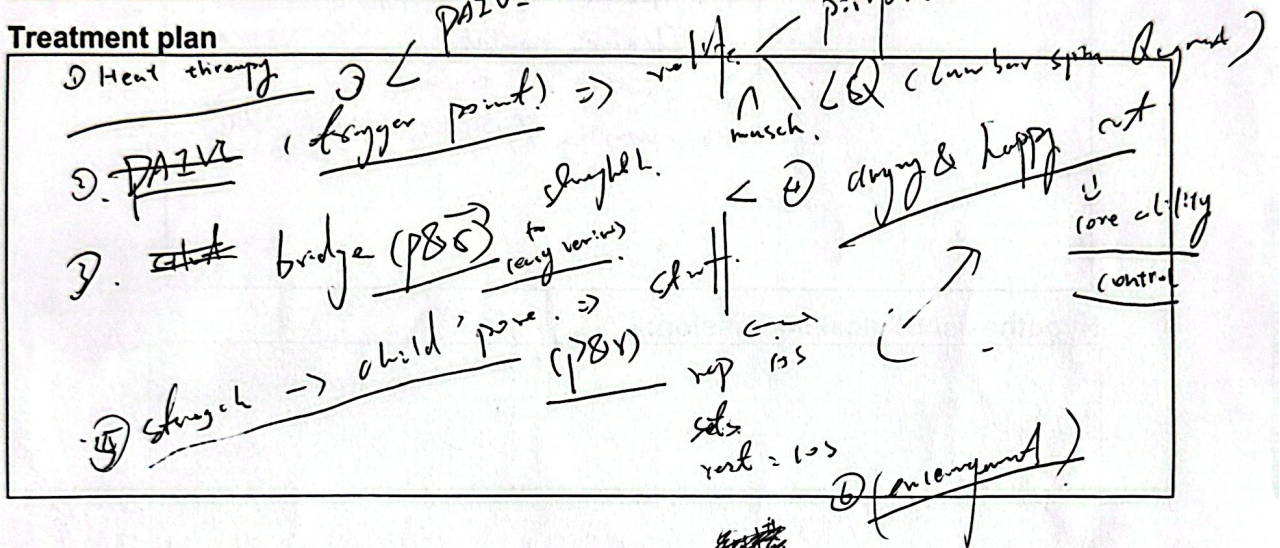
Hypothesis/ Clinical impression:

Clinical Reasoning (Why)?

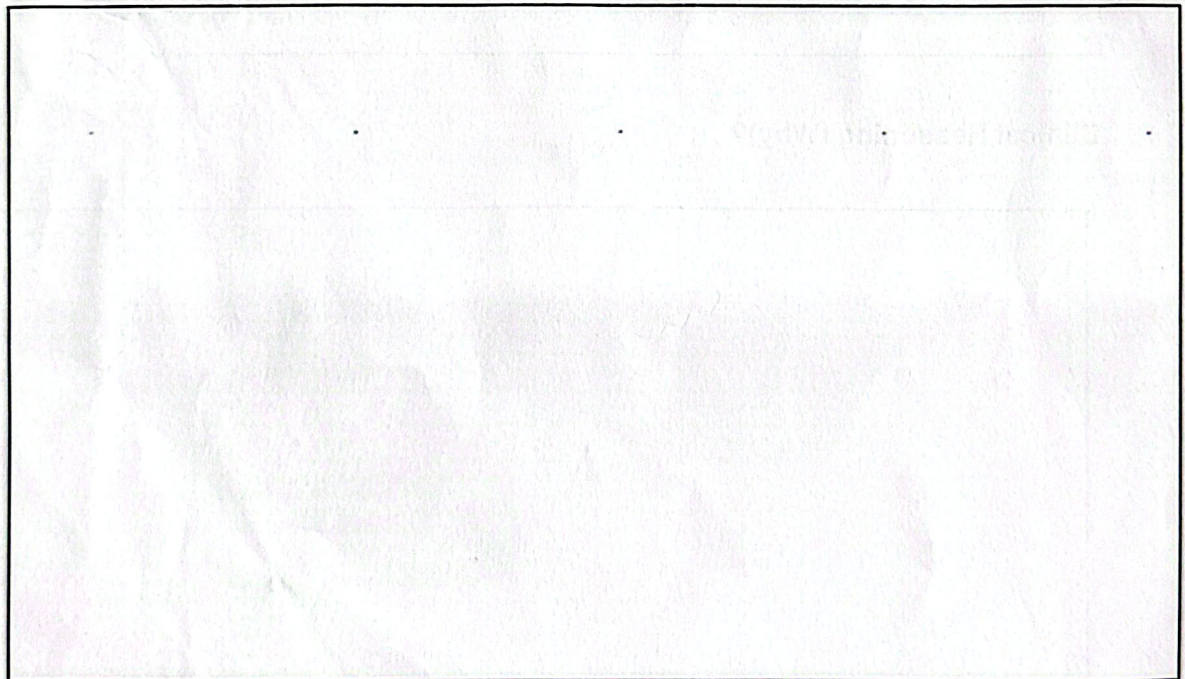
Problem list / Contributing Factors.



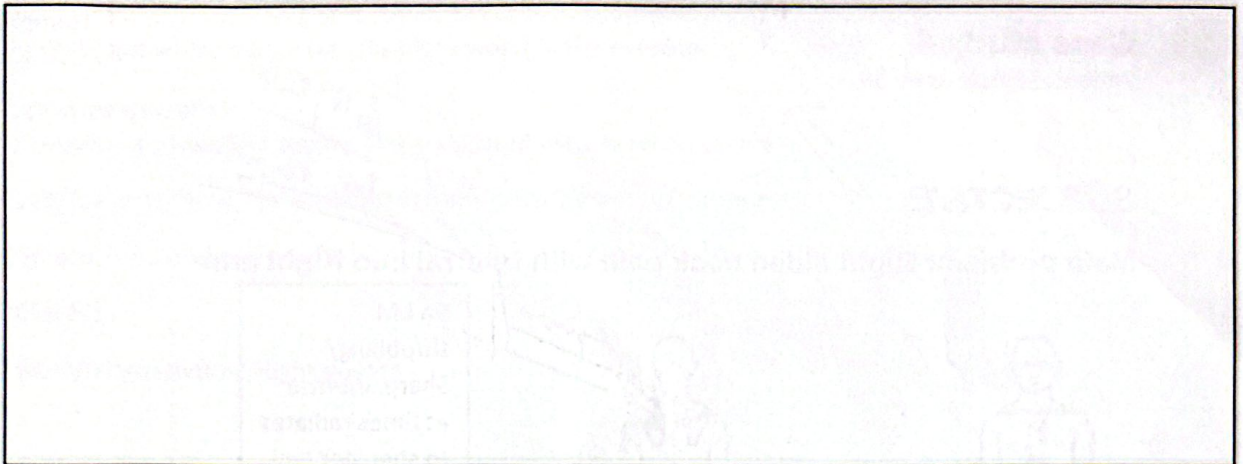
Treatment plan



Clinical Reasoning – Why??



Goals



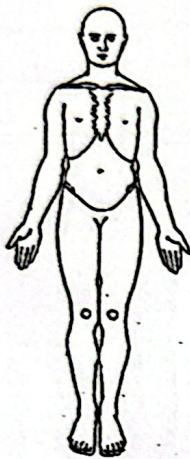
Case study 4

Patient: Female aged 58

Thoughts?

SUBJECTIVE

Main problem: Right sided neck pain with referral into Right arm



Pa I.M
throbbing/
Sharp. Intense
at times radiates
to shoulder and
associated with
Pb
NRS 4-8/10.

Pb I.M pain and
tingling RIGHT
thumb

moderate-high
Sensory

maybe nerves problem

Aggravating factors:

Looking up with head back – immediate, but mild Pa

Turning to the right Pa and sometimes Pb

Sitting in neutral Pa – 60 mins

Lifting /carrying items > ~10kg – 30-60mins Pa

Housework / DIY / decorating – 4hours Pa+ Pb

Easing factors

Changing position – 5-10 mins

Resting lying on back – 5 mins if only Pa, 30mins if Pb

Night

Difficulty getting comfortable - likes L sd.ly

Unable to sleep on R as will wake patient. Firm mattress and 2 pillows

irritating
mod

AM

Often stiff and achy takes several mins to get going

Tends to get worse during the day often worst in the evening.

Special questions

Intermittent p+n RIGHT thumb (Pb) associated with Pa when severe

°gait/°balance/°trips/°LL weakness/°clumsiness/°dexterity/°clumsiness

Left arm is normal

°D's/ N's

°fever/fatigue/wtloss/night sweats

Thoughts on special questions/ red flags? 

History of present condition:

4 month history of sudden onset of Right sided neck and shoulder pain. Pain started after a busy weekend lifting boxes into new house with partner. Went to GP after ~2months who advised to keep moving and take naproxen for 2 weeks. Pain persisted so referred to physio.

Patient reports concern about the problem as they still have a lot of decorating to do in their new house and the problem seems to flare up every time they do something. Patient is frustrated with this ++.

Never had any physio before and no previous problems with neck.

Past medical history

Fit and well

°Thyroid/°Heart/°BP/°Inflammatory Arthritis/°Asthma

°Epilepsy/°Diabetes/°Surgery/°Cancer

Drug history

Naproxen (now finished the course).

Contraceptive pill

°Long term steroids, °Anticoagulants

Investigations

Nil

Social history

Lives with partner in their new house

Hobbies

Normally goes to body pump, Pilates. Avoidant of these classes now as doesn't want to flare her symptoms up. Also plays Badminton and hurts to serve and hit overhead smash

Occupation

Graphic designer – desk based works from home at least 50% of time.

Thoughts on yellow flags?

Beliefs/concerns

Told it might be a trapped nerve. She thinks she needs a scan

Expectations

Get rid of the pain

Have a scan

Help to return to classes

ANALYSIS

Hypothesis/ Clinical diagnosis:

S mod - high 4-8/10
I mod
N

Any precautions/contraindications for Ax?

OBJECTIVE

Observation

Slumped with slight forward head posture and mild Tsp/Csp kyphosis in sitting – correctable with facilitation to neutral.

Normal gait and balance

• signs of muscle atrophy

Csp ROM

	AROM	Active over pressure
Flex	Chin to chest, °p	Sl. Pa EOR
Ext*	½ range, ↑Pa and b	NT
L Rotn	80°, °p	√√
R Rotn*	40° ↑Pa and b	NT
L SF	20°, °p	30°
R SF	15°, stiff, sl Pa	NT

Shoulders

Full AROM, ° pain

upper limb test ⇒ test median nerve

Muscle tests

Craneo-Cervical Flexion test (CCF in lying) unable to sustain 10 sec inner range hold, recruiting SCM.

头伸屈外, Hold ⇒ deep neck flexors ability

Scapular setting – poor awareness, unable to hold 10 neutral posture.

Upper traps tight, worse on right

Special tests

Spurling's test – Right side positive reproduces Pa and Pb

Neuro

Dermatomes: sl. dulled to light touch over C6 on right, but can still discriminate sharp and blunt

Myotomes: normal 5/5 throughout

Reflexes: Biceps, Triceps, Brachioradialis all equal and present, Babinski / clonus / Hoffman's all normal

Neurodynamics

ULTT1 +ve on Right, lacks 20° from full elbow extension

↑ Pa, Pb and p+n, sensitised with Left SF Csp

Palpation/PAIVMs

Palpation of median nerve trunk right arm tender

PA C5 & 6 IV jt- sl. sore. Unilateral PA C6 facet joint on right ↑ Pa++

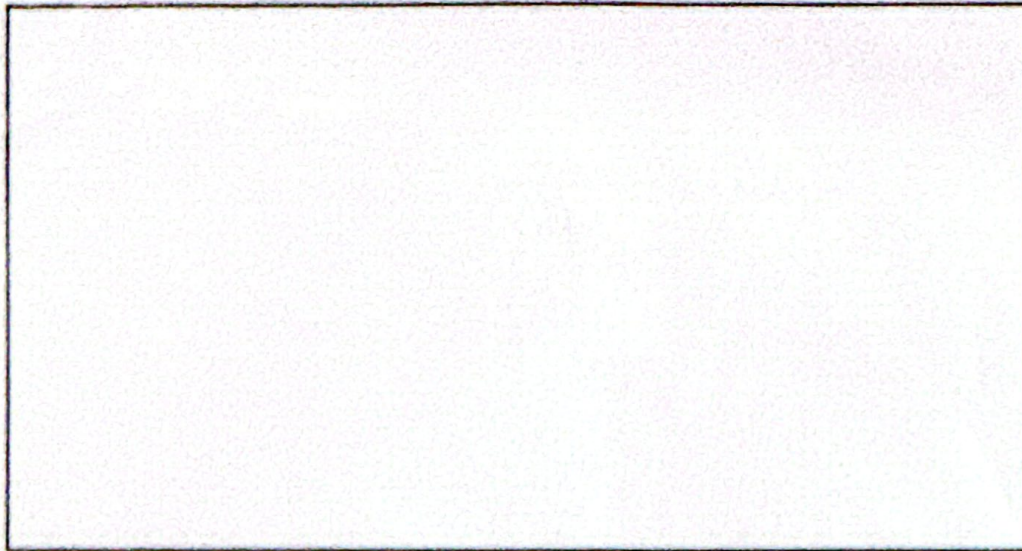
Tight upper traps R > L

ANALYSIS

Facet joint pain.
nerve root pain C5-C6 ⇒ vertebral pain.
myogenic. component

Hypothesis/ Clinical impression:

Clinical Reasoning (Why)?



Problem list / Contributing Factors.

neck and shoulder Levator scapulae tight. upper trapezius tight.	Twist joint pain / Sore. Upper trapezius Levator scapulae tight. Altered posture Pain on rotation - side Reduced strength Lighting therapy Upper limb test
---	--

Treatment plan

- ① Heat therapy - mobilize, relax, warm for therapist to do further work
- ② trigger point
- ③ manual therapy for twist joint
- ④ passive stretch for

levator scapulae
upper trapezius

}

back self stretch exercise
passive by therapist
actively using hand by the self
- ⑤ strength training for deep neck muscle
- ⑥ neuro mobilization
- ⑦ posture education

massage (relax)

<
- ⑧ encouragement

Demost-ne → 运动链 → 肩 → 肘 → 手 → (下肢) → 骨盆
为外例

frankel's test → 呼吸功能检查

Clinical Reasoning - Why??

下肢肌力, 用力 (ask patient to lift)

levator → pendulum

lumbo spine: side flexion (与右大腿左右扭) / figure 4 measurement

Crook lying

straight leg raise

length test (pumping test)

Ham - Cervical test

Slump test → Goals
sheer & gapping test

Caenstons test

intra compression

侧躺压颈椎

PSIA
E
F
→ point
→ 颈椎

pelvic ring → around curved
元. 大腿 2 points
L2 L3 L4 L5

low limb myotome →

Upper limb Tension Test: check (pic), 用腿推上臂, elbow extension

Scapular stability → prone lying (进入姿势) (H/A arm)

observe pelvis → PSIS / ASIS / iliac spine

palpate T4 → T3 / T12

cervical ROM & OP →

Reflex → Achil / knee

elbow 有两个 → key: compare two sides

L2-L4 → iliac crease → L3-L4

Muscle strength → 5/5 → 3/5 grade / not strong

Thoracic spine ROM → 双手交叉 → 颈椎 & 肩关节 fixed.

Trapezius test → 把头前侧下压